Genomic adaptation of *Achromobacter* spp. in patients with primary cilliary dyskinesia

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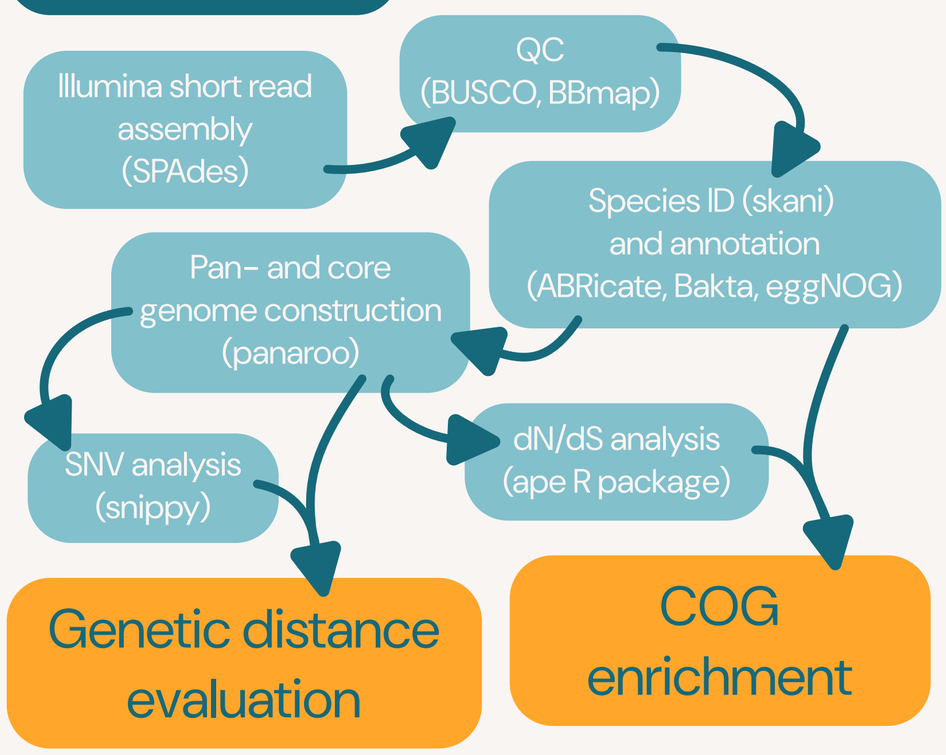
Introduction

Over the past decades increasing **pathogen resistance to antibiotics** is posing a threat to public health [1]. One genus of such opportunistic pathogens is ***Achromobacter* spp.** that is **inherently resistant** to antibiotics, increasingly causing infections in people with primary ciliary dyskinesia (PCD), leading to **often fatal complications**. Understanding how pathogens adapt to the host environment is **essential for improving and developing new treatment strategies**.

Methods



Dataset

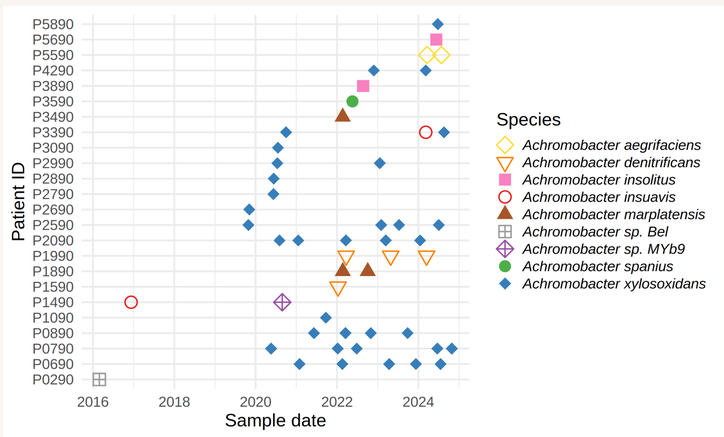


Fig. 1: Timeline of patient sample collection and Achromobacter species identified

Samples were collected between 2016 and 2024 from **24 people with PCD** at Rigshospitalet, Copenhagen, and were identified as belonging to **nine distinct species** of *Achromobacter*, with ***A. xylosoxidans* being most prevalent**. Notably, two patients (P1490 and P3390) were infected with **more than one species** during the sampling period.

Results

Gene presence–absence and SNV distances

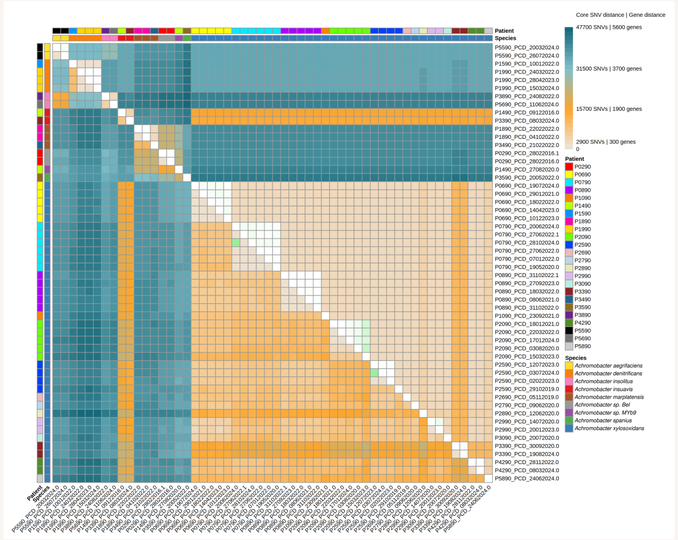


Fig. 2: Pairwise genomic distance heatmap. The lower triangle shows pangenome gene presence–absence differences, and the upper triangle shows core genome single nucleotide variant (SNV) distances.

Within-patient samples generally showed **low genomic distances**. Distances within species were higher but remained lower than those observed between species. ***A. xylosoxidans* and *A. insuavis*** showed the **lowest interspecies distances**, while other species displayed larger distances to one another. A notable exception was patient P3390, who was **co-infected** with *A. xylosoxidans* and *A. insuavis*. In this case, the patient's *A. xylosoxidans* isolates showed distances to other *A. xylosoxidans* strains **comparable to those observed between *A. xylosoxidans* and *A. insuavis***.

Known AMR factors

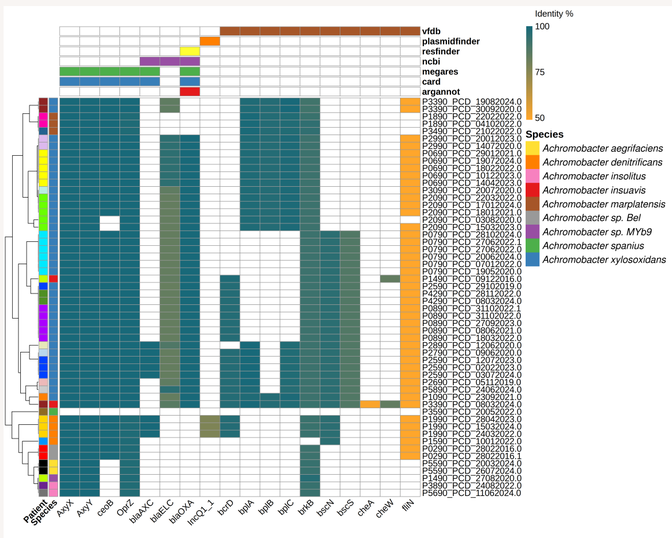


Fig. 3: Antimicrobial resistance factor search results across seven databases. Color intensity in each tile represents the nucleotide identity match percentage.

A search in seven antimicrobial resistance (AMR) factordatabases revealed presence of genes responsible for:

- AxyXY-OprZ **efflux pump** system
- CeoB inner membrane RND transporter
- bla family **beta-lactamase** enzymes, blaOXA being effective against carbapenems [2]
- **Virulence** factor BrkB that helps **evade** the host's **complement system**
- Flagellar **motor switch protein** FliN
- **Chemotaxis** che family proteins (*A. insuavis* only)
- Bsc and Bcr **type III secretion** systems
- **Beta-lactam resistance** Bpl family proteins (although their orthologs are more likely)
- **Plasmid replication** protein IncQ_1 (P1990 only)

Genes under positive selection

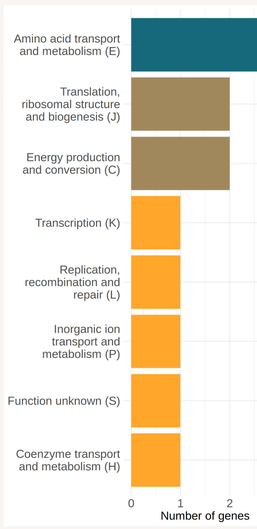


Fig. 4: Frequency of COG functional categories among genes showing multiple dN/dS > 1 in more than one sample.

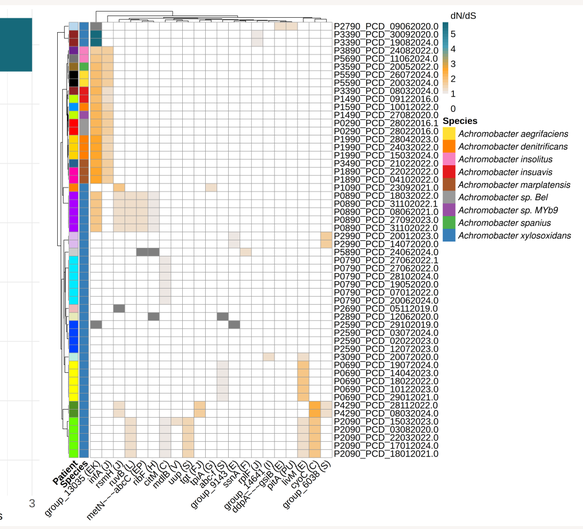


Fig. 5: dN/dS ratios across all genes and samples. Only genes with dN/dS > 1 in at least one sample are shown. COG categories are indicated in brackets.

A dN/dS analysis against a consensus core genome identified positive selection in **22 genes in at least one** sample, and in **12 genes in at least two** samples. COG enrichment showed that selected genes were mainly associated with **amino acid** transport and metabolism (E), **protein synthesis (J), energy production (C)**, as well as DNA **transcription (K), replication and repair (L)**, and transport and metabolism of ions (P) and coenzymes (H).

References

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Main takeaway

Achromobacter spp. display **antimicrobial resistance and virulence** factors, with **active selection in DNA maintenance and protein synthesis** genes driving adaptations to the respiratory environment in PCD patients via **mutation and regulation**, that could be used as **potential targets** when developing novel therapeutic strategies.

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